

Power and Spares

Master Control

File: power.kicad_sch

File: bcontroller.kicad_sch

PCI Bus

ATA Interface

File: pcibus.kicad_sch

File: IDE.kicad_sch

Agnus Multiadapter

HID and User Ports

File: Agnus-Multi.kicad_sch

File: hid.kicad_sch

Chip RAM Controller

Audio

File: cpu.kicad_sch

File: audio.kicad_sch

PCI Bridge

Video

File: pci-bridge.kicad_sch

File: video.kicad_sch

Motorola Bus Logic

Chip Memory

File: motorola_bus_logic.kicad_sch

File: chip_memory.kicad_sch

Serial

Clocks, ROM, USB

File: serial.kicad_sch

File: clocks.kicad_sch

Floppy and Parallel

PCI Bus Logic

File: floppy.kicad_sch

File: pci_bus_logic.kicad_sch

100 = PCI BUS
200 = CHIPSET
300 = VIDEO
500 = AUDIO
600 = POWER
700 = CHIPSET RAM
800 = VOLTAGE SHIFTER
900 = ATA

BUDDY

DESIGNED WITH KICAD REV 8

Sheet: /
File: AmigaPCI Mainboard.kicad_sch

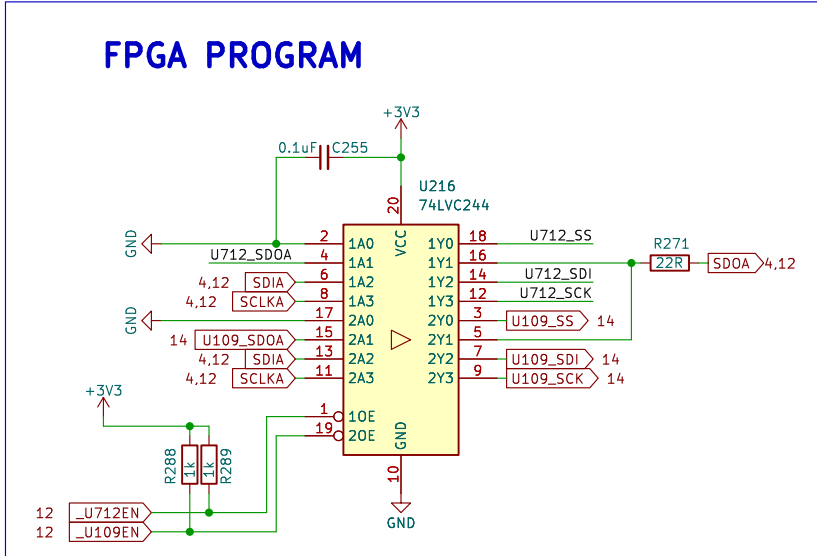
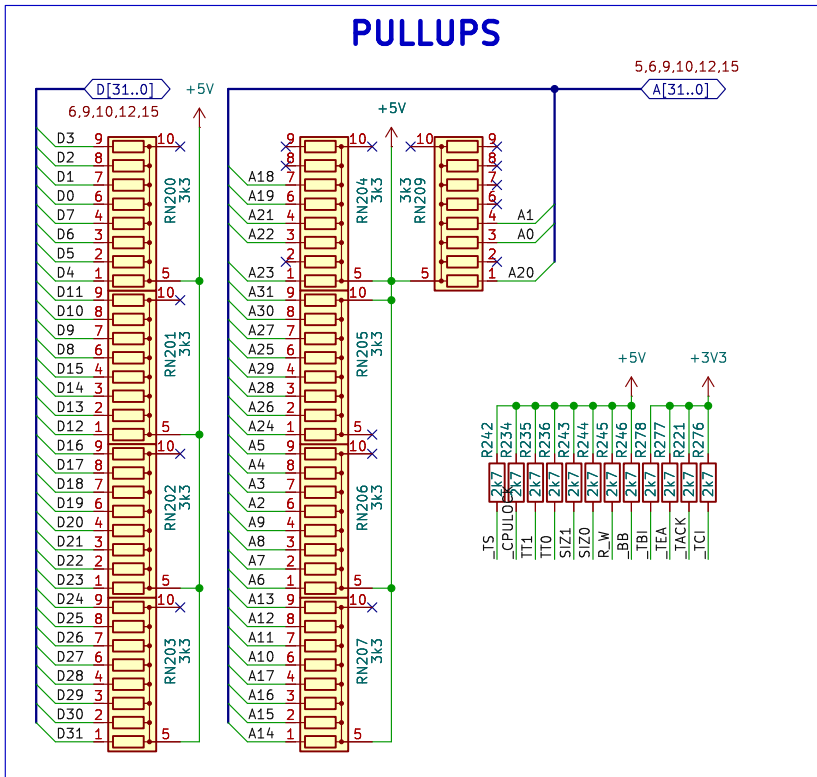
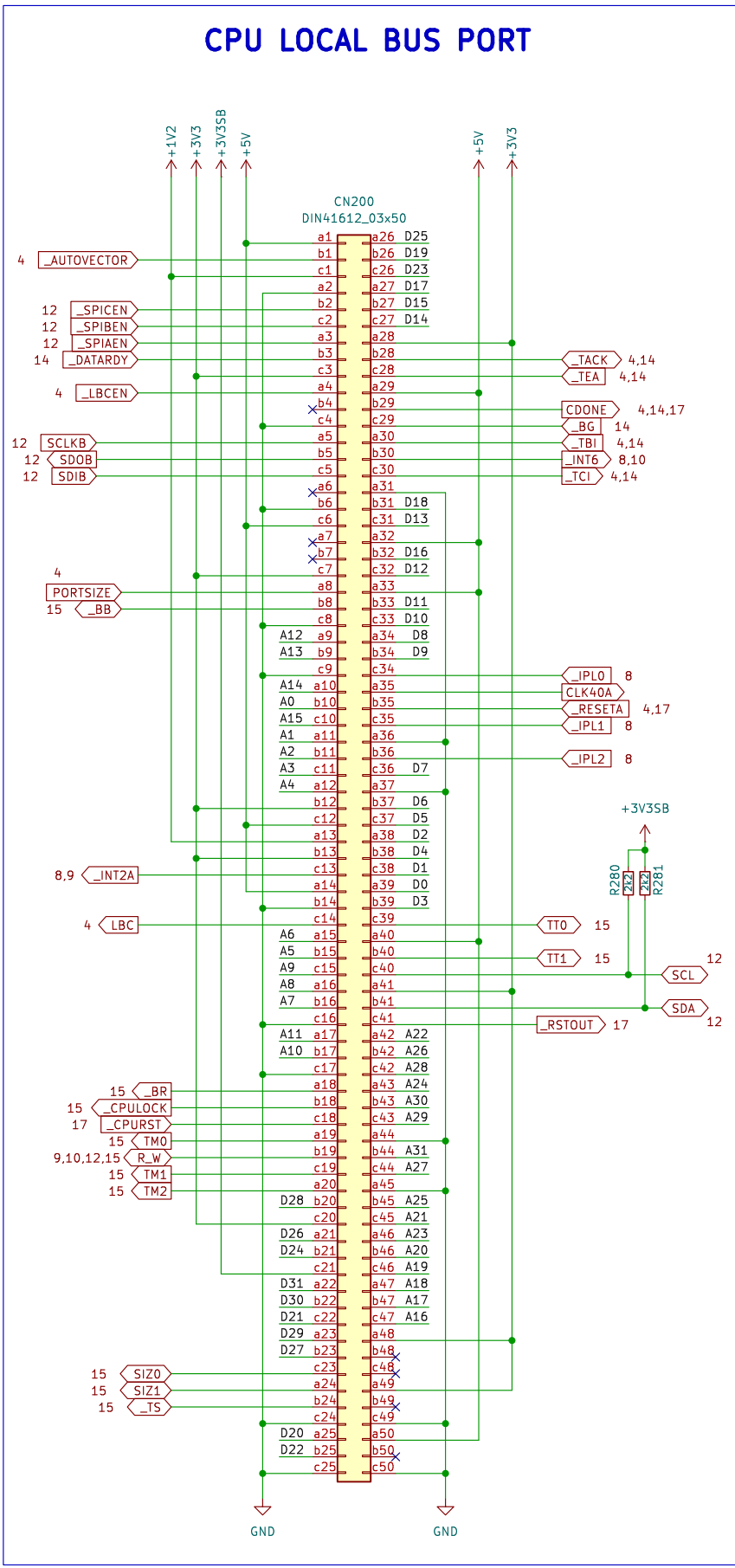
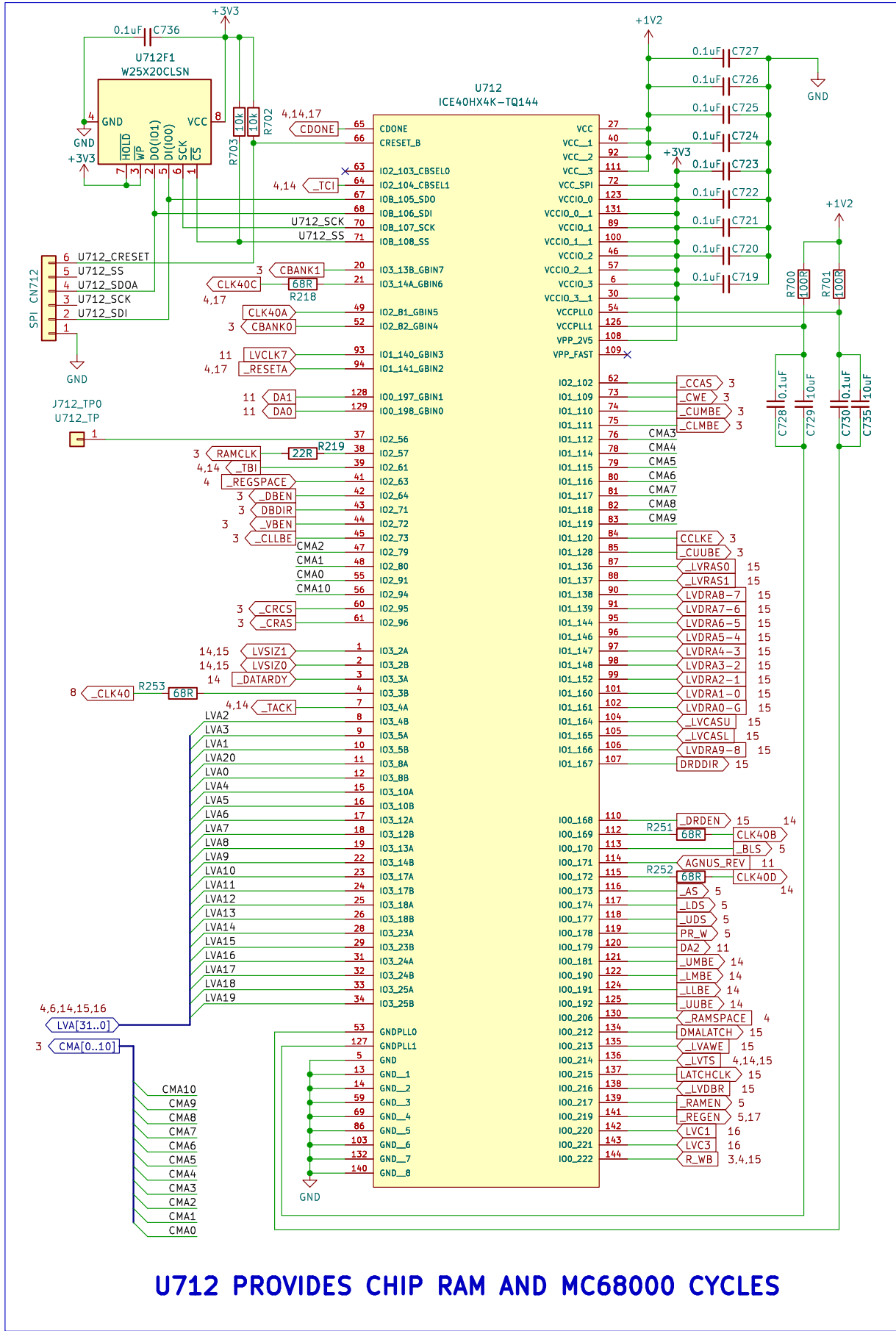
Title: AMIGA PCI

Size: USLedger	Date: 2026-04-27
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Rev: 7.0

Size: 05Ledge1	
KiCad E.D.A. 8.0.9	

id: 1/17



CHIP RAM CONTROLLER

Sheet: /Chip RAM Controller/
File: cpu.kicad_sch

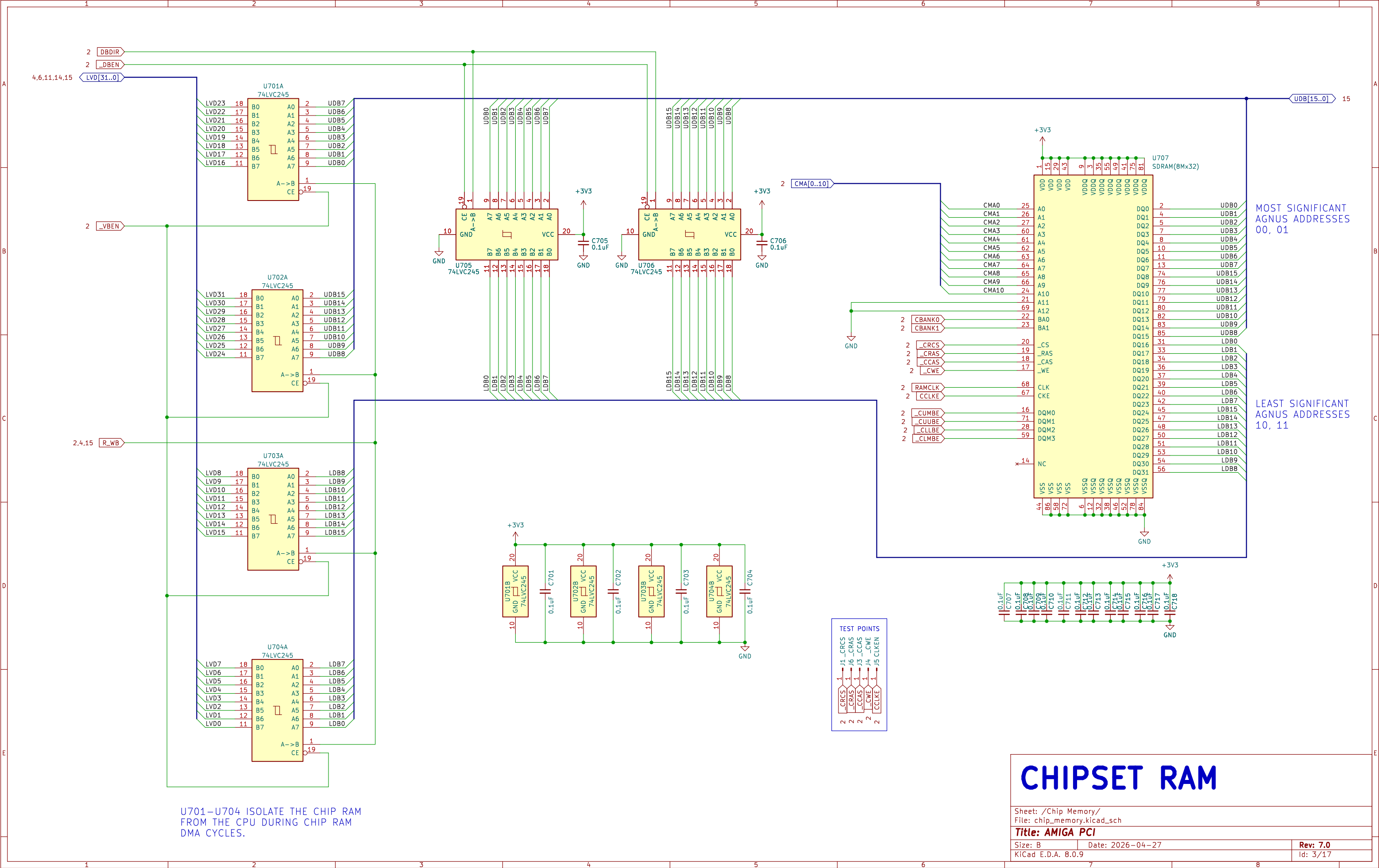
Title: AMIGA PCI

Size: B Date: 2026-04-27

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Rev: 7.0

Id: 2/17





Sheet: /Master Control/
File: bcontroller.kicad_sch

Title: AMIGA PCI		
Size: B	Date: 2026-04-27	Rev: 7.0
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SUPPORTS 8372A (ALL PART NUMBERS) AND 8375 THE FOLLOWING PART NUMBERS:
390544-01 AND COMPATIBLE (390544-02, 390554-01, 390554-02, 318069-12, 318069-13) AND 318069-10, 318069-11

JUMPERS MUST BE CONFIGURED BY AGNUS PART NUMBER

U229M
74CBT16233

13A 21 13B1 36 13B2 35 34

+5V

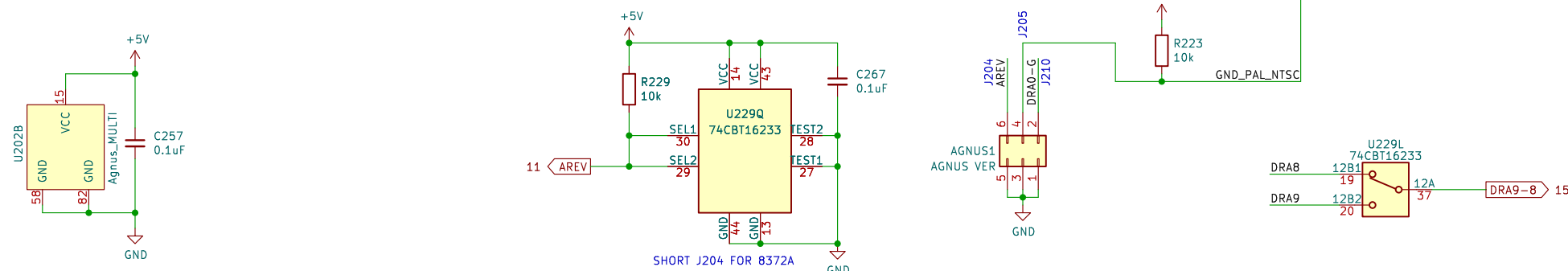
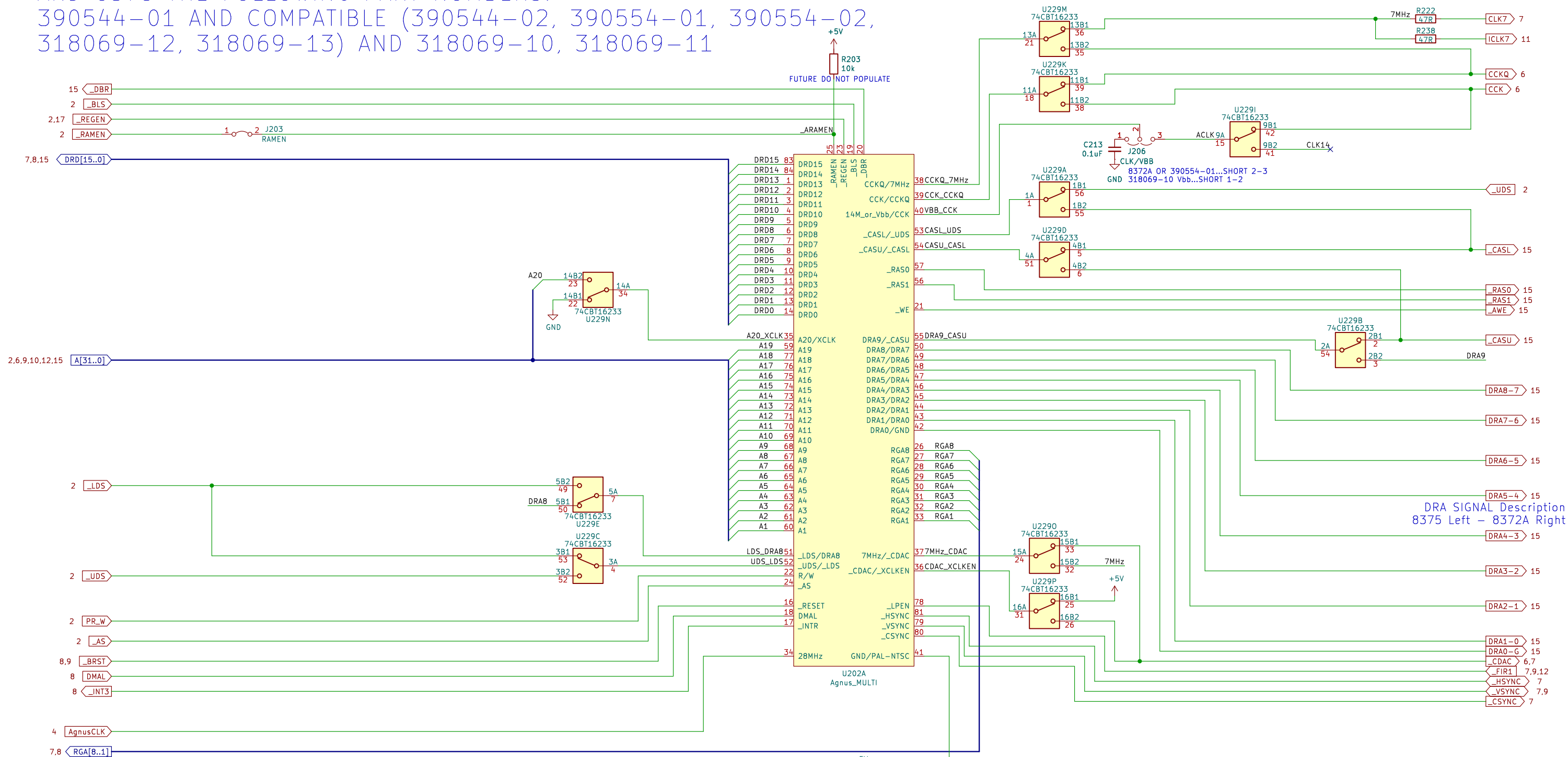
7MHz

R222 47R

R238 47R

CLK7 7

ICLK7 11



Agnus 8375 and 8372A Multiadapter

Sheet: /Agnus Multiadapter/
File: Agnus-Multi.kicad_sch

Title: AMIGA PCI

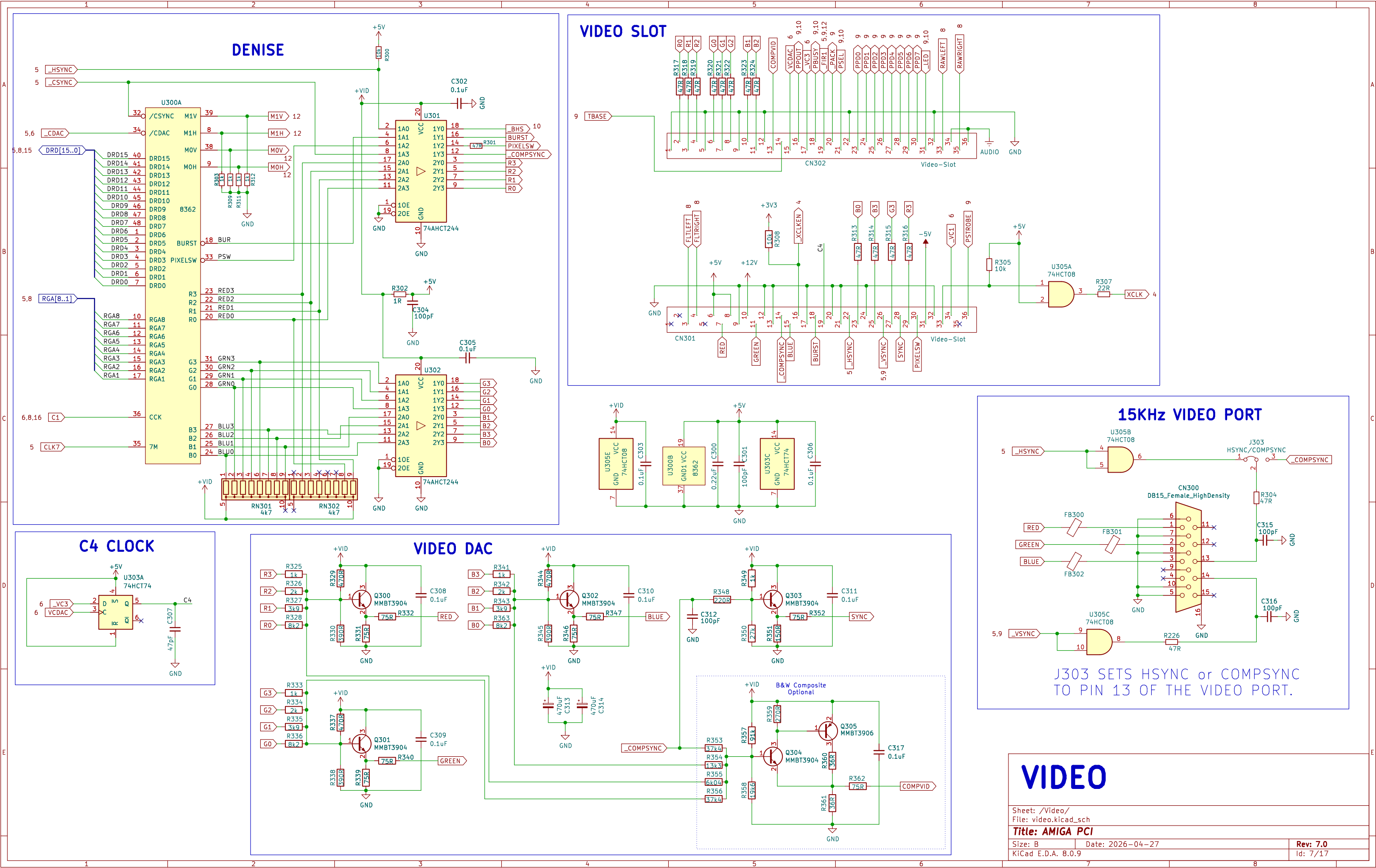
Size: B	Date: 2026-04-27
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Rev: 7.0

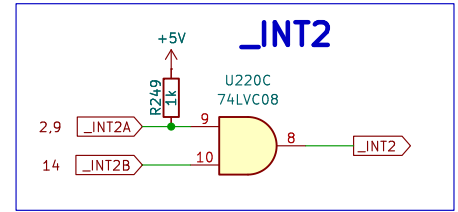
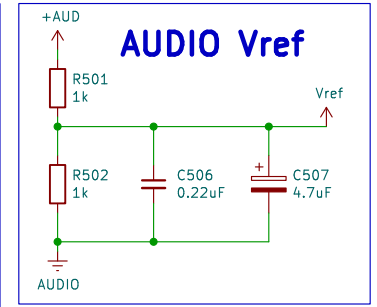
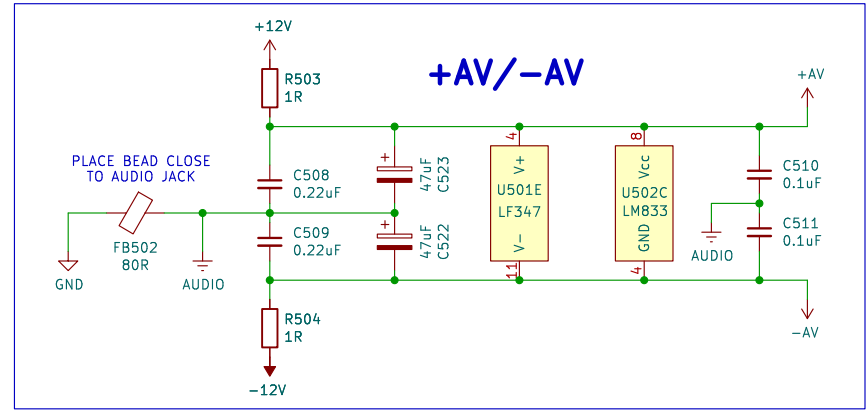
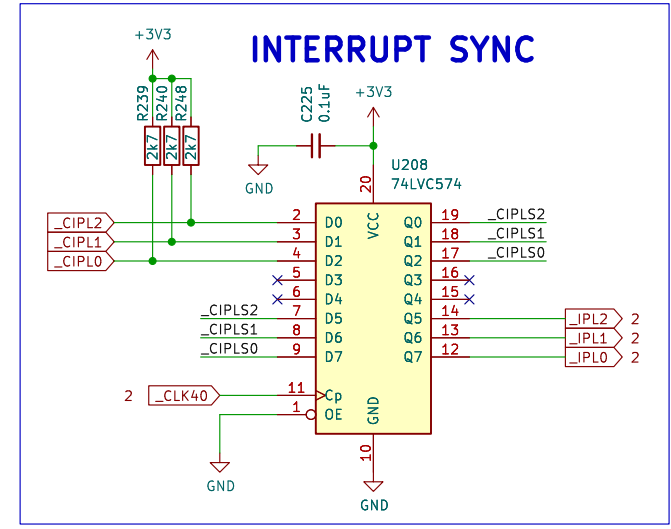
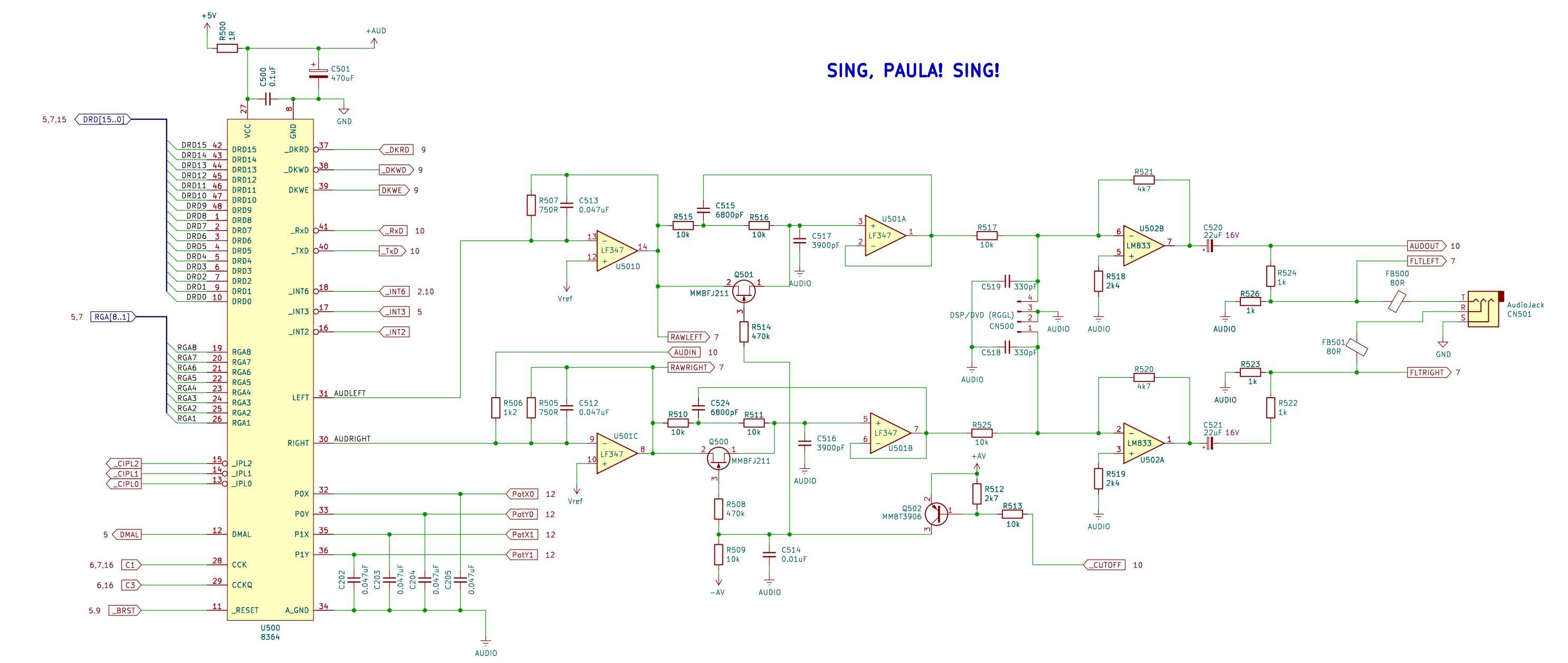
KiCad E.D.A. 8.0.9

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Id: 5/17

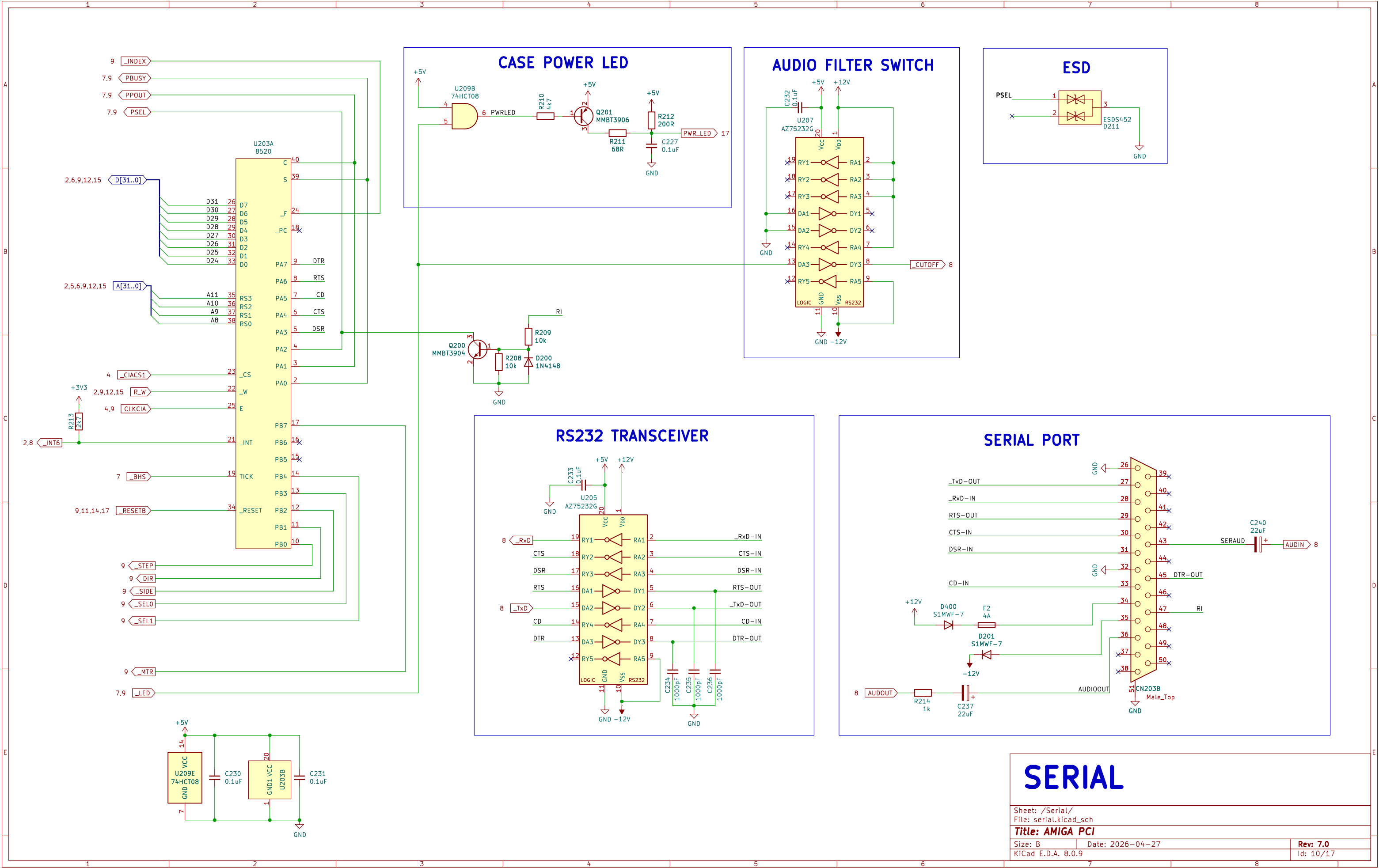


SING, PAULA! SING!



PAULA/AUDIO

Sheet: /Audio/ File: audio.kicad_sch		
Title: AMIGA PCI		
Size: B	Date: 2026-04-27	Rev: 7.0
KiCad E.D.A. 8.0.9		Id: 8/17



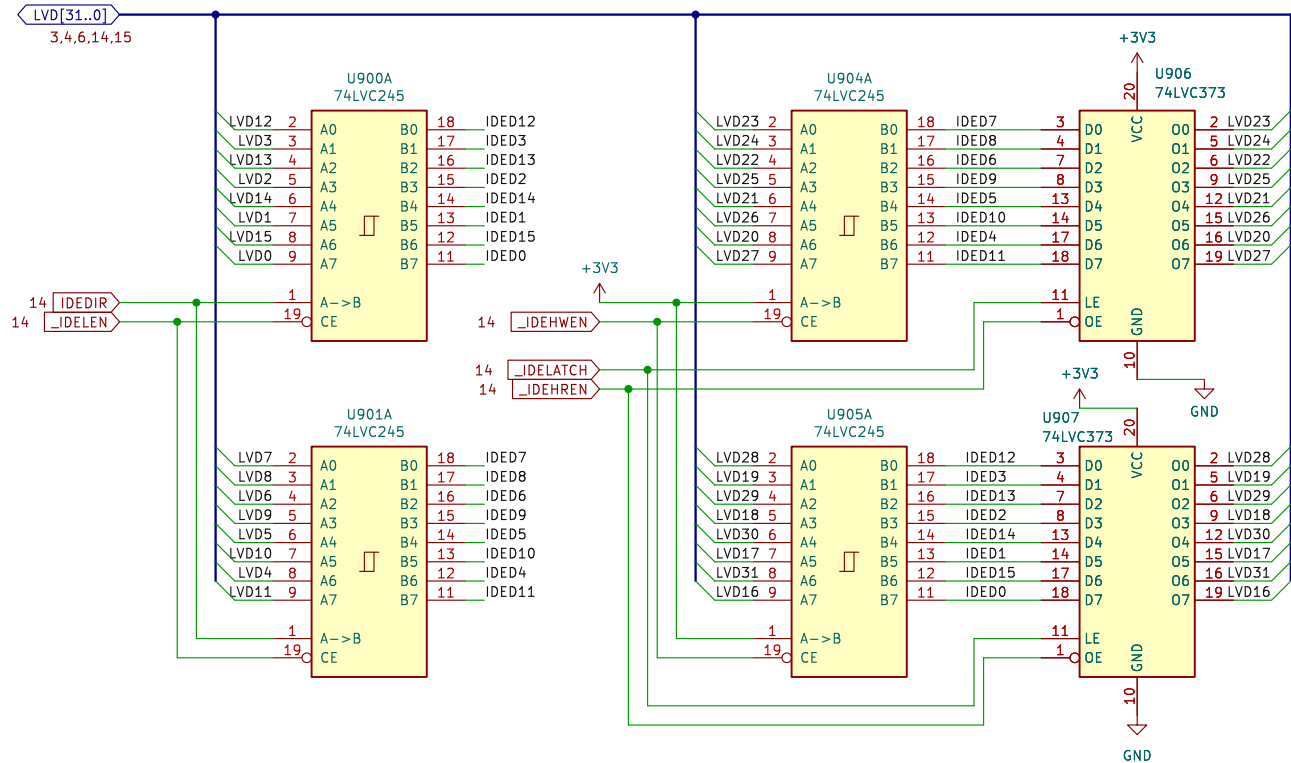
DATA BUFFERS

74LVC245 ENABLED FOR FIRST WORD DURING WRITES.
DATA ALWAYS FLOWS A->B.

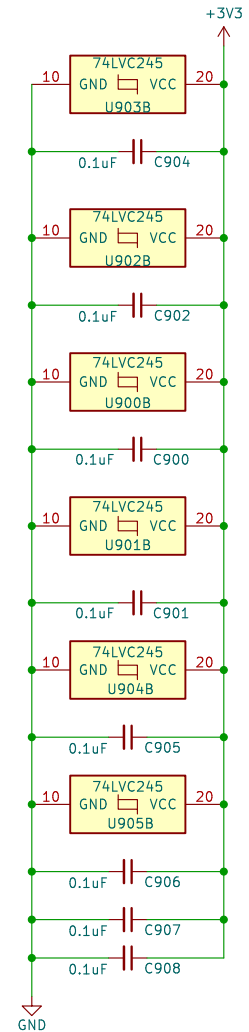
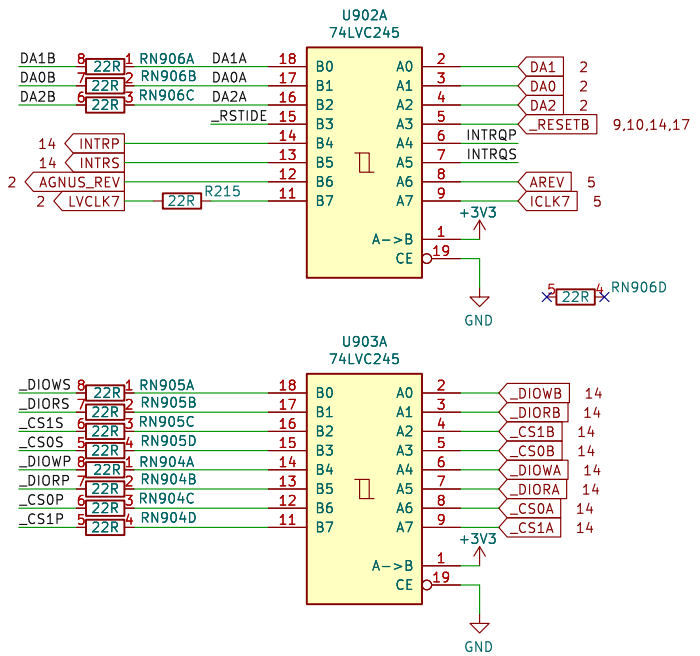
74LVC373 LATCHES DATA FOR FIRST WORD DURING READS.

DURING WRITE CYCLES, ALL DATA IS REAL-TIME.
FOR THE FIRST WRITE COMMAND, D16..31 IS ENABLED.
FOR THE SECOND WRITE COMMAND, D0..15 IS ENABLED.

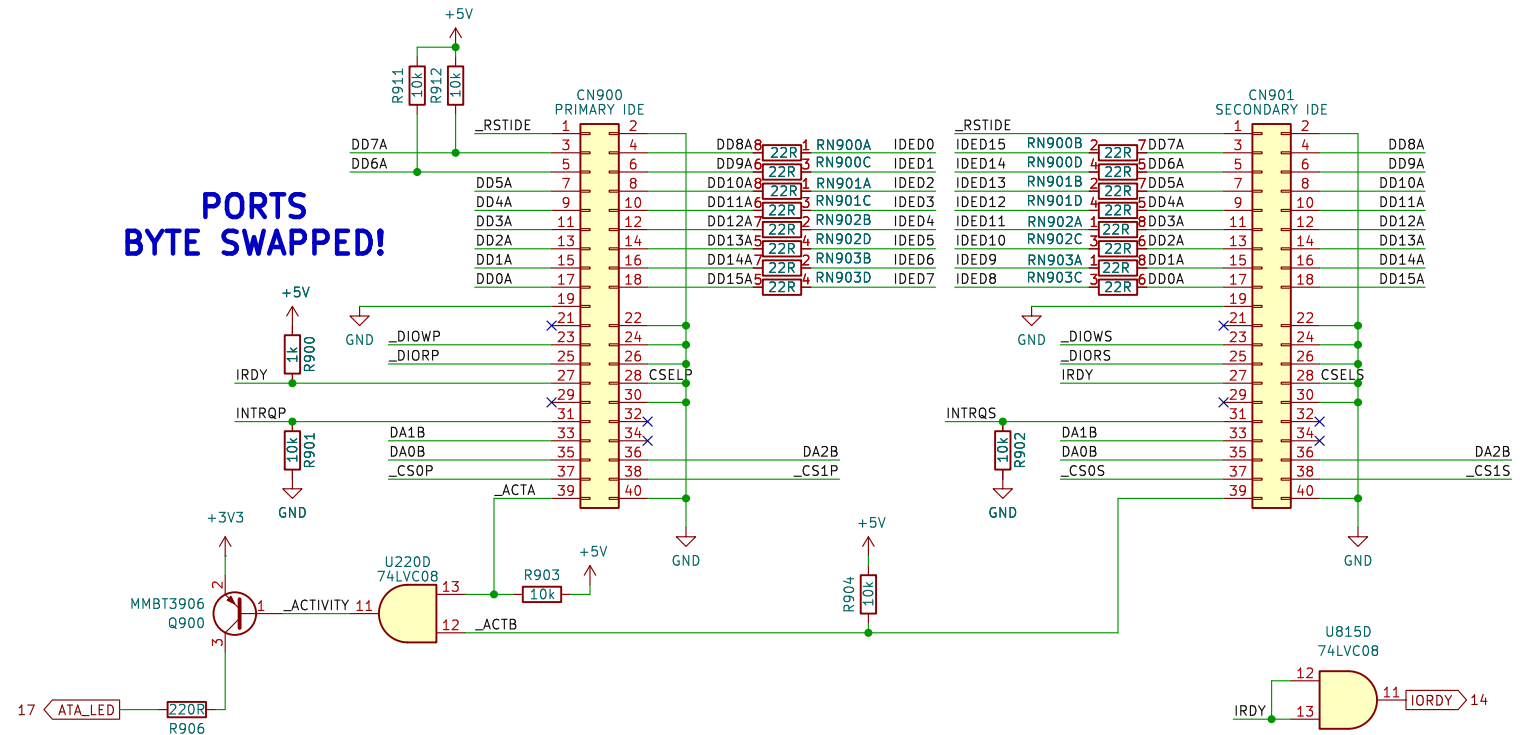
DURING READ CYCLES, D16..31 IS LATCHED AFTER THE FIRST READ COMMAND FROM THE IDE DEVICE. THE SECOND READ FROM THE ATA DEVICE DRIVES D0..15 LIVE. THE DATA IS THEN LATCHED BY THE MC68040.



BUFFERS AND TERMINATION



PORTS BYTE SWAPPED!



ATA INTERFACE

Sheet: /ATA Interface/
File: IDE.kicad_sch

Title: AMIGA PCI

Size: B

Date: 2026-04-27

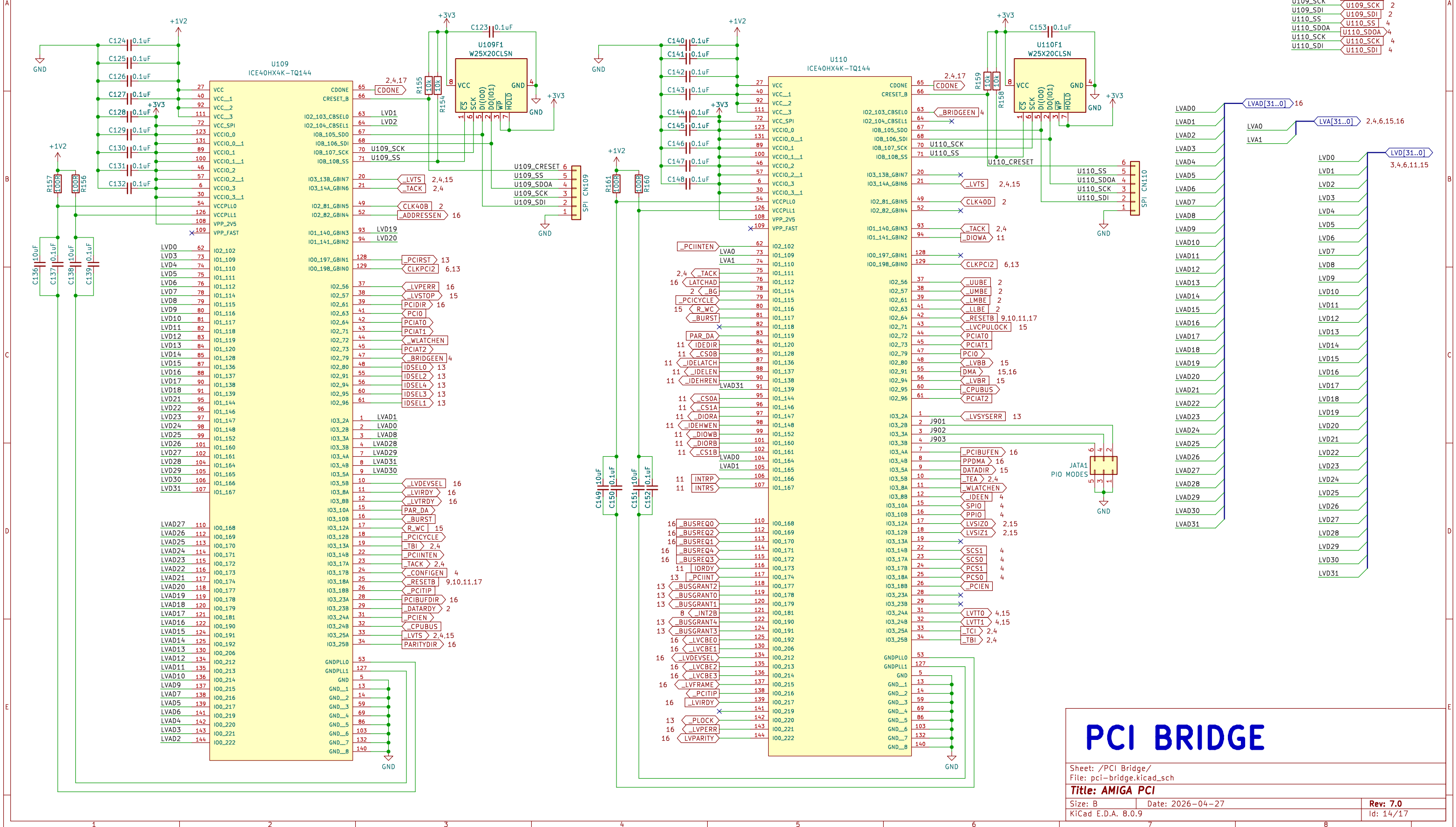
Rev: 7.0

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11/17

U109: BUFFERS AD/D/A BUSES, AD PARITY CALC, BRIDGE REGISTERS

U110: BUS ARBITOR, INTERRUPT HANDLING, PARITY ERROR DETECTION, ATA CONTROLLER



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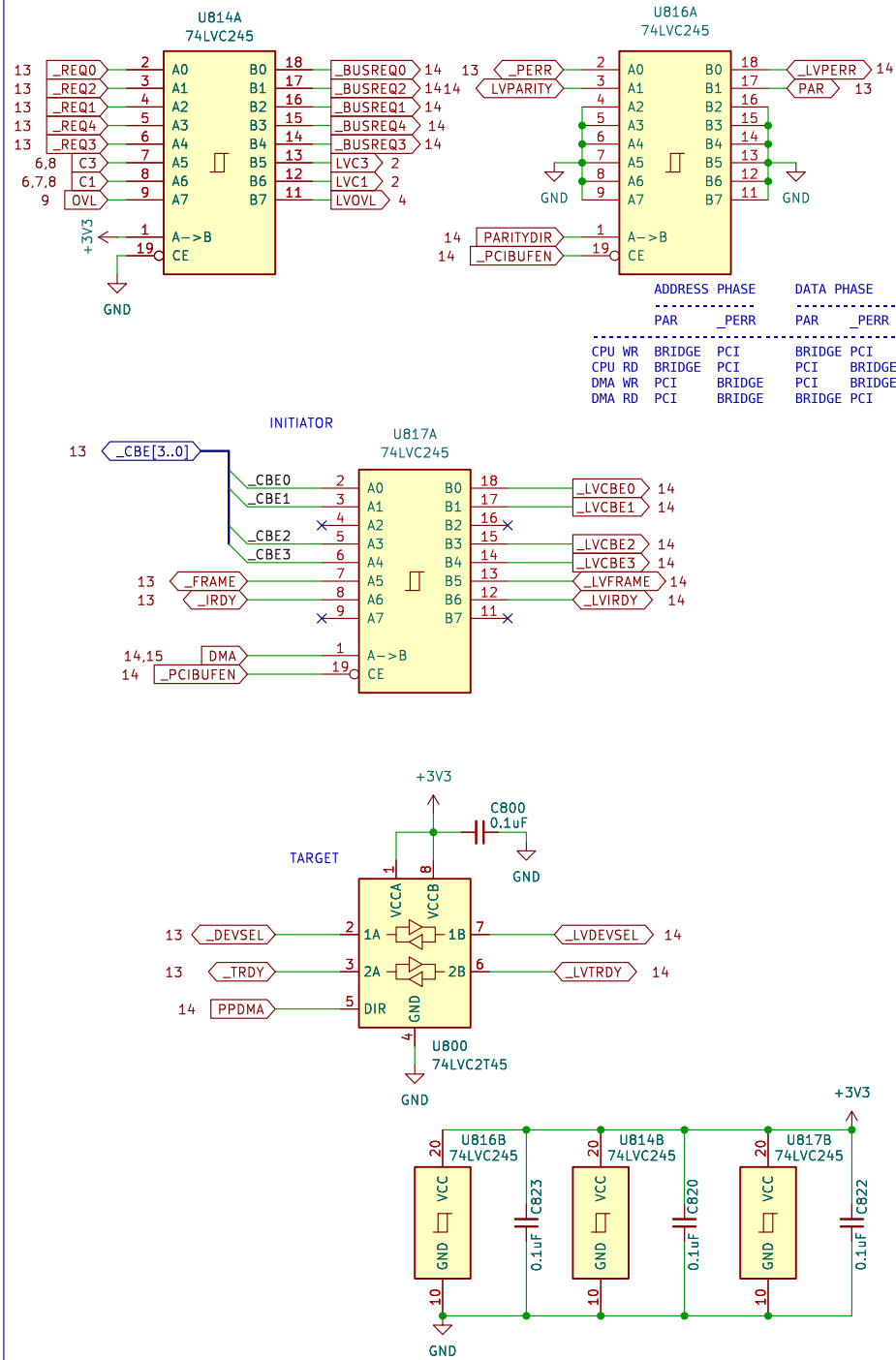
Size: B	Date: 2026-04-27	Rev: 7.0
KiCad E.D.A. 8.0.9		Id: 15/17

Title: AMIGA PCI

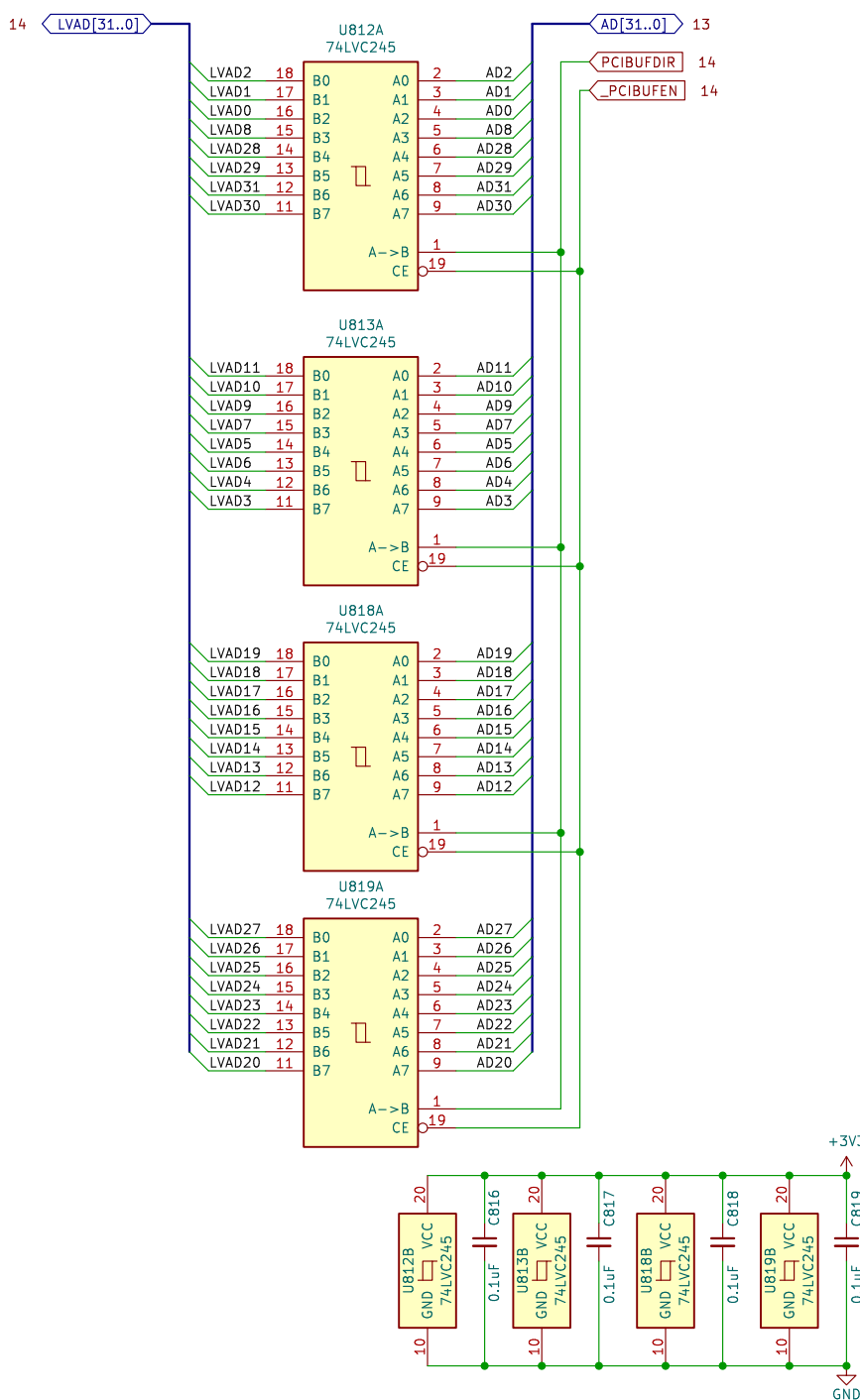
KiCad E.D.A. 8.0.9

d: 15/17

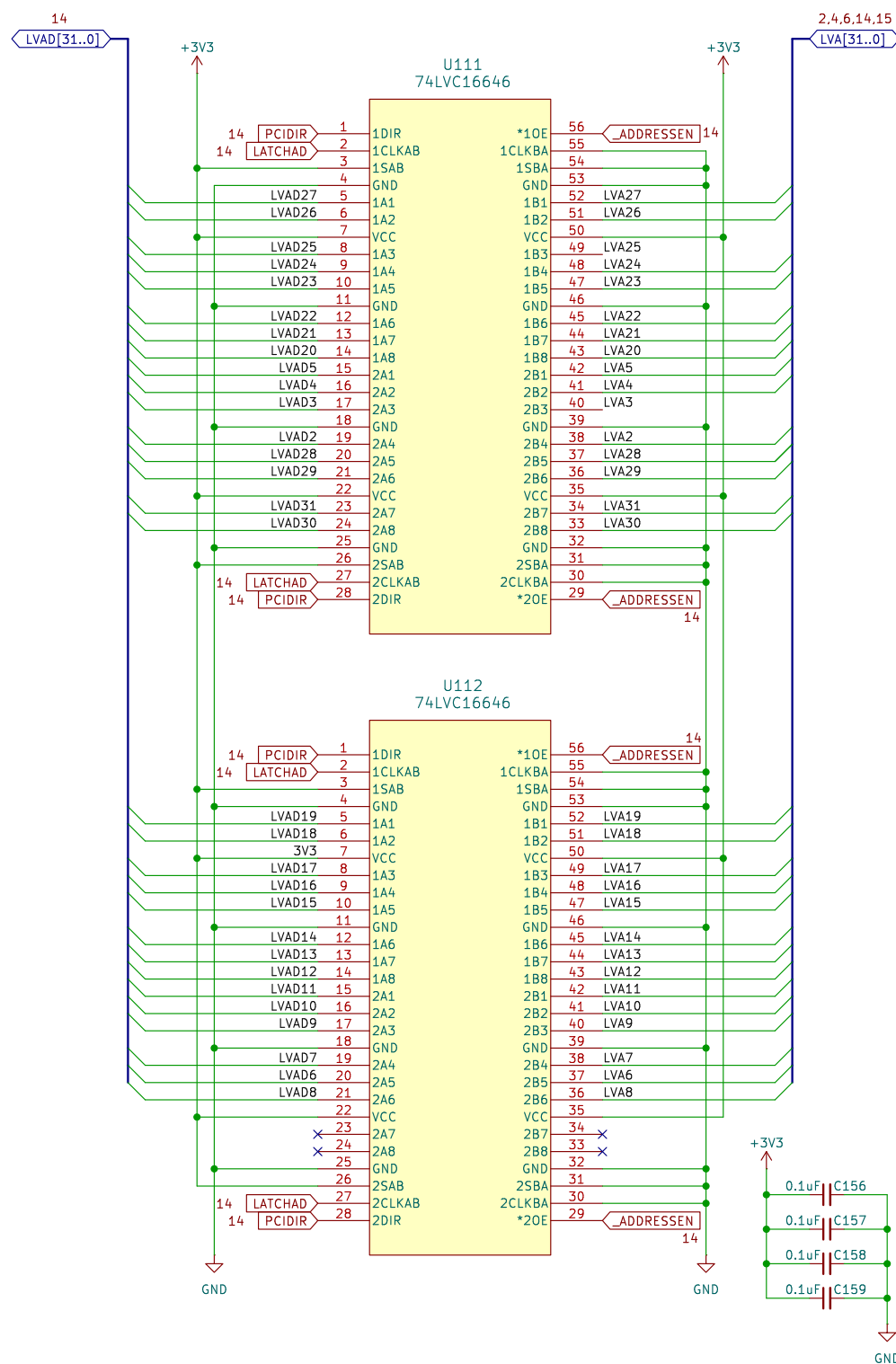
PCI BUFFERS



AD BUS LEVEL SHIFT



PCI ADDRESS LATCH



PCI BUS LOGIC

Sheet: /PCI Bus Logic/

File: pci_bus_logic.kicad_sch

Title:

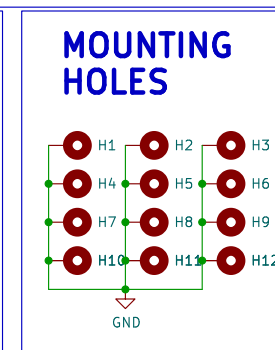
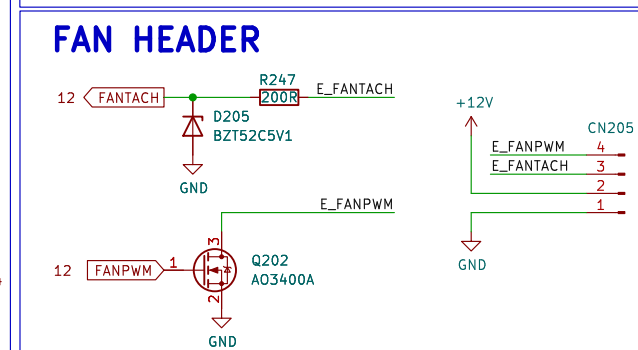
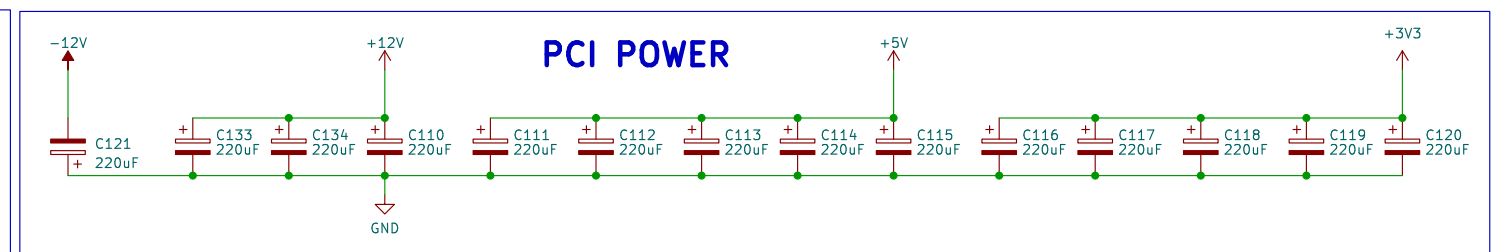
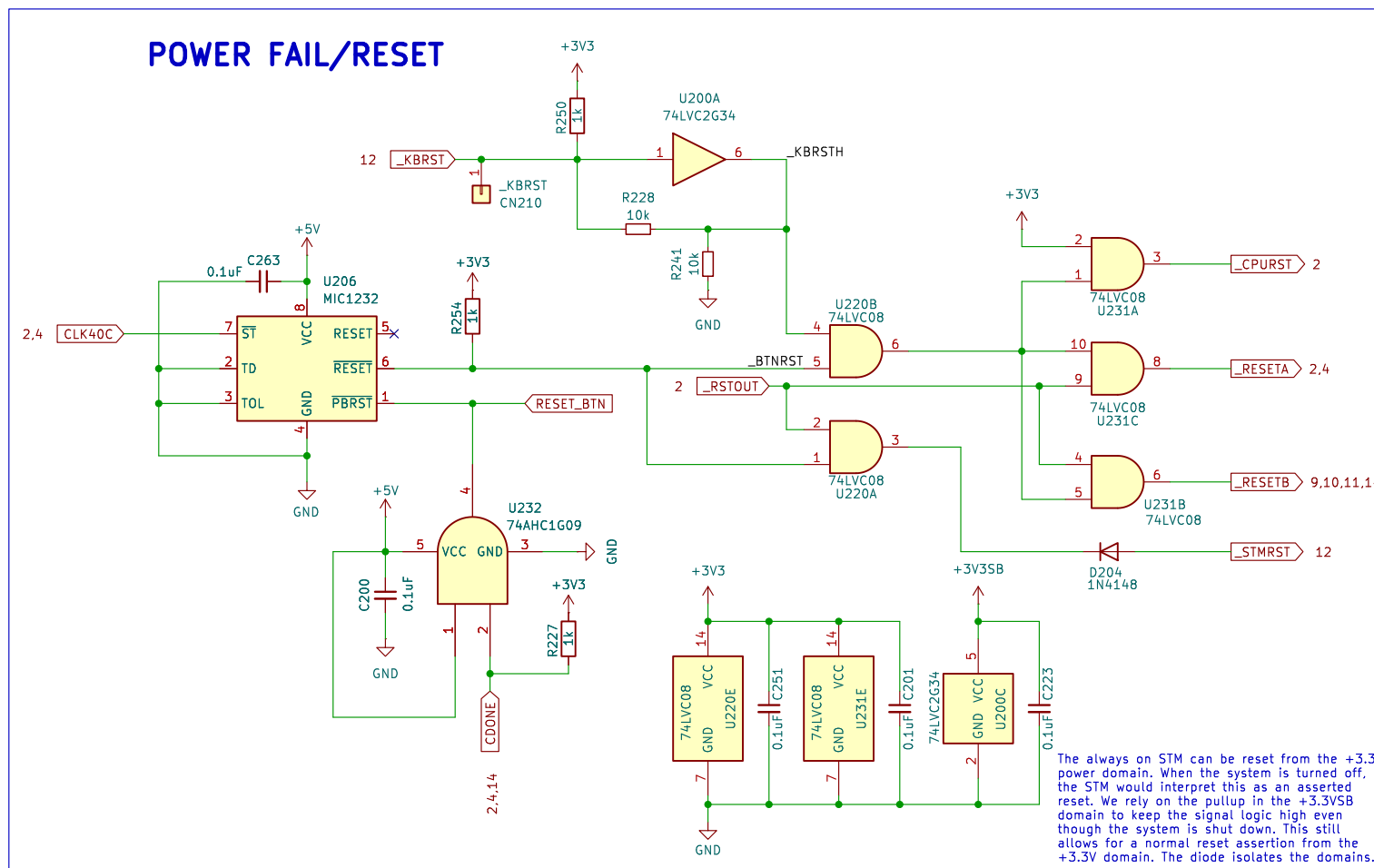
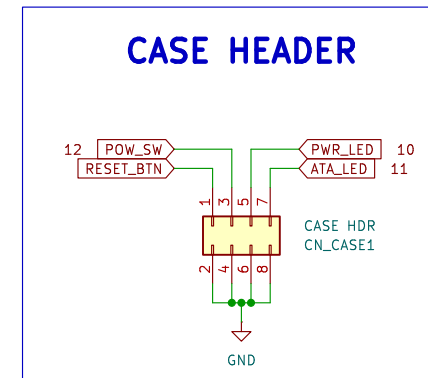
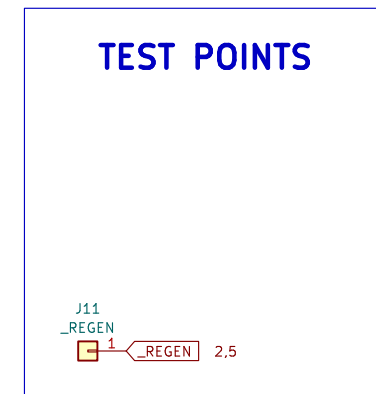
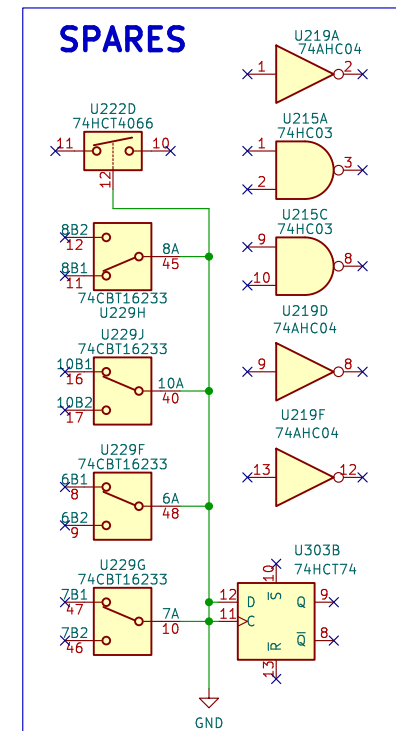
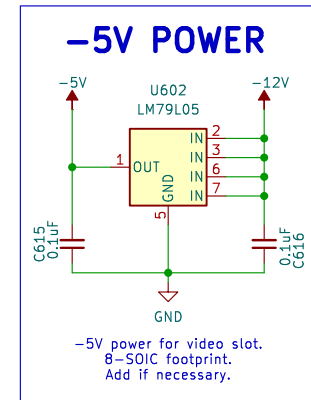
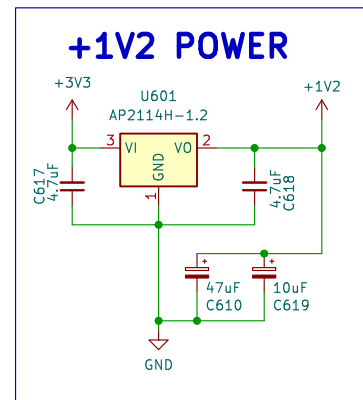
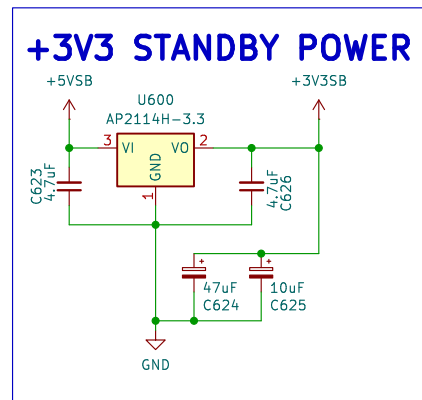
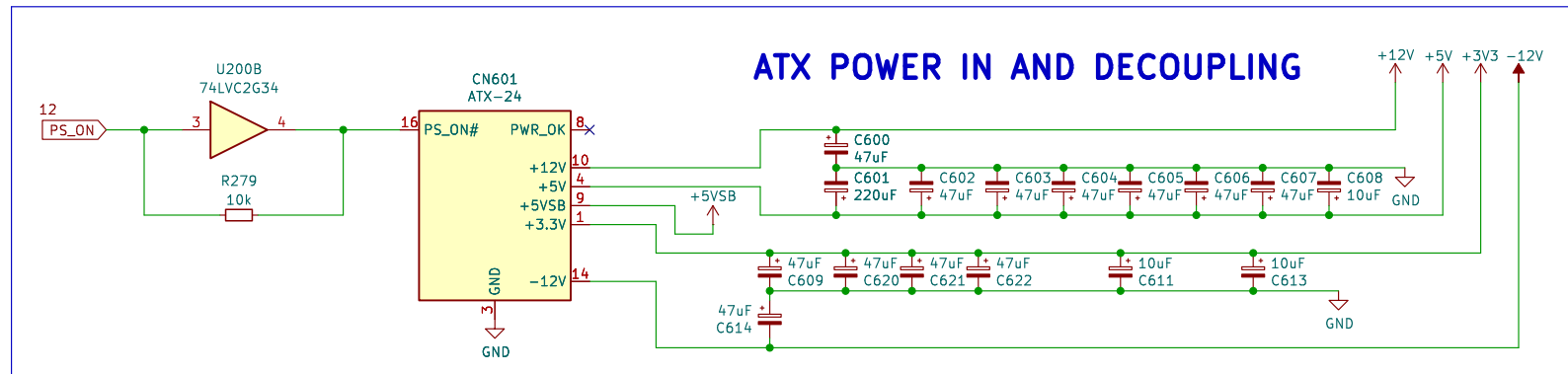
Size: B

Date: 2026-04-27

Rev: 7.0

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Id: 16/17



The always on STM can be reset from the +3.3V power domain. When the system is turned off, the STM would interpret this as an asserted reset. We rely on the pullup in the +3.3VSB domain to keep the signal logic high even though the system is shut down. This still allows for a normal reset assertion from the +3.3V domain. The diode isolates the domains.

POWER/SPARES

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Title: AMIGA PCI

Size: B

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Rev: 7.0

KiCad E.D.A. 8.0.9

Id: 17/17